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(54) **METHODS AND APPARATUS FOR A LEAD
FIXATION DEVICE**

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(57) **ABSTRACT**

A lead fixation device for attaching a deep brain stimulation lead within in a burr hole in a human skull. The lead fixation device may comprise a mounting plate and an insertion collar extending from a lower surface of the mounting plate and configured to be received in within the burr hole in the human skull. The mounting plate may comprise a mounting flange, a first arm and a second arm. The mounting flange and the first arm may comprise a static mounting hole while the second arm may comprise an adjustable mounting hole. The second arm may be movable to an operating position when the second arm is compressed toward the first arm, the inner walls of the first and second arms are substantially parallel, and the mounting screw is tightened within the adjustable mounting hole when the second arm is in the operating position.

20 Claims, 11 Drawing Sheets

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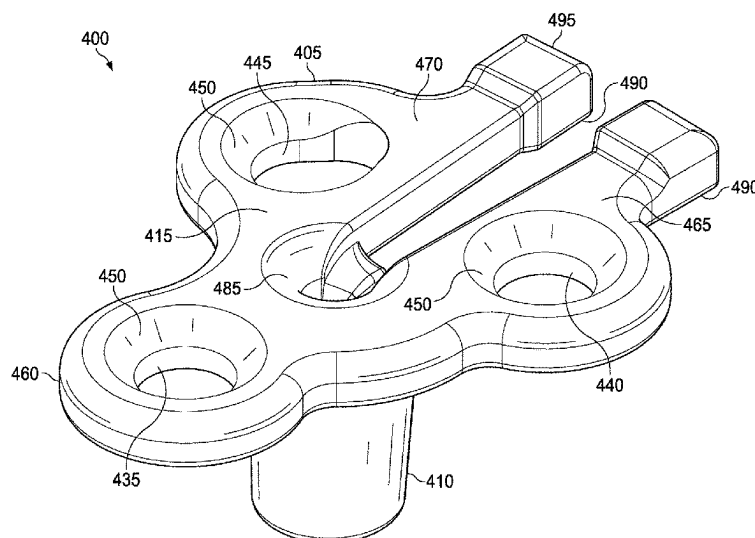
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continuation-in-part of application No. 15/715,895,
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on Nov. 6, 2018.

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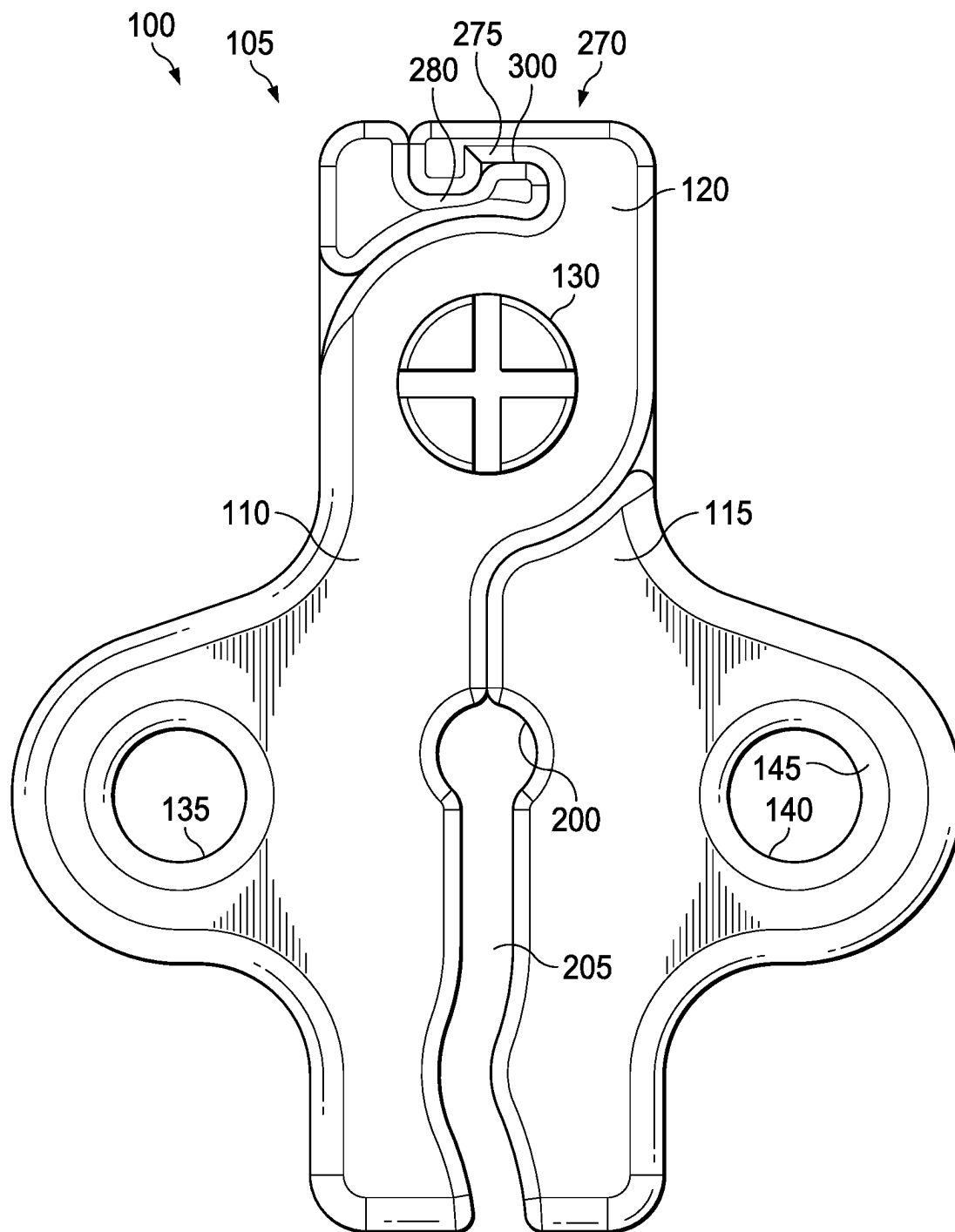


FIG. 1

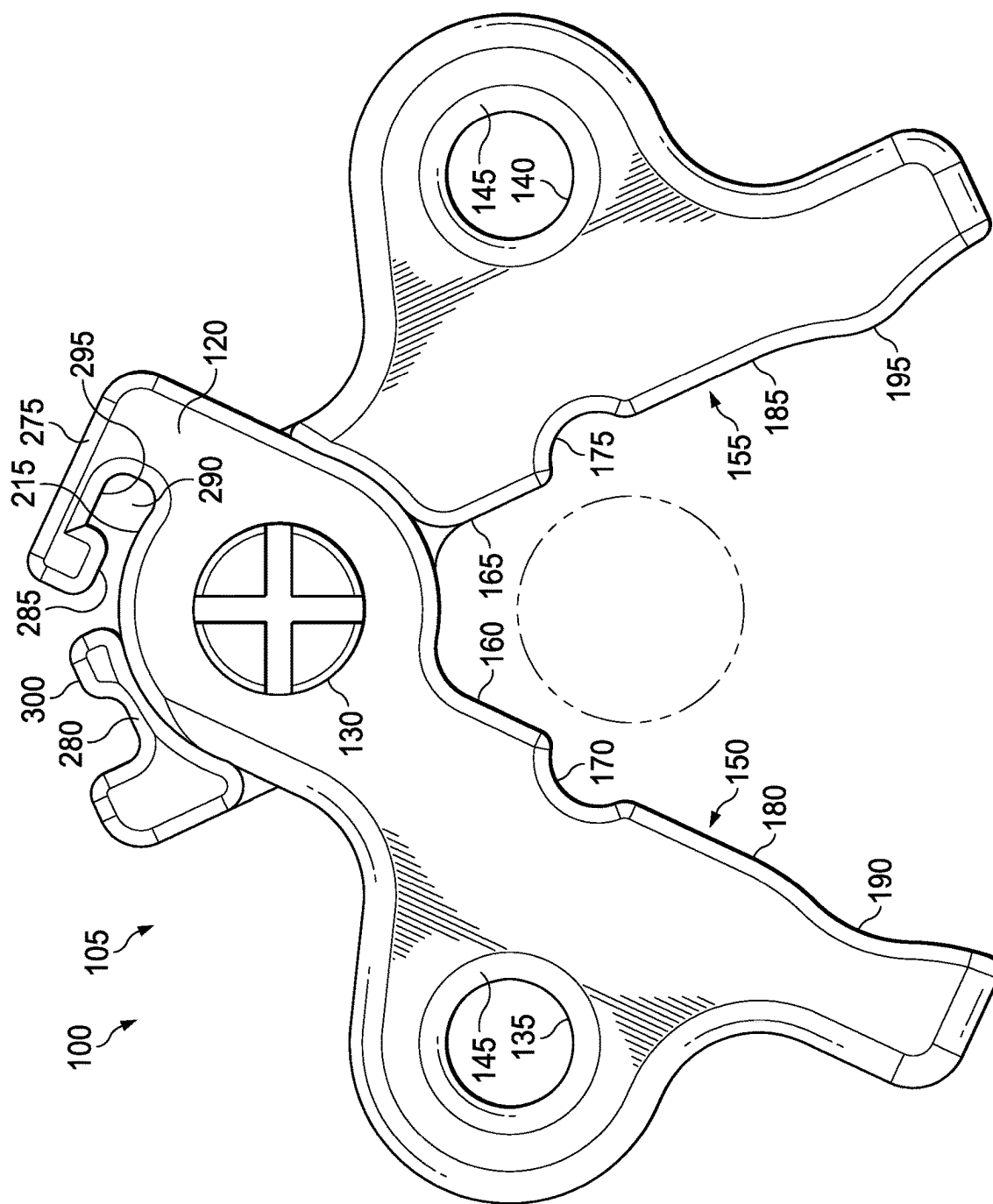


FIG. 2

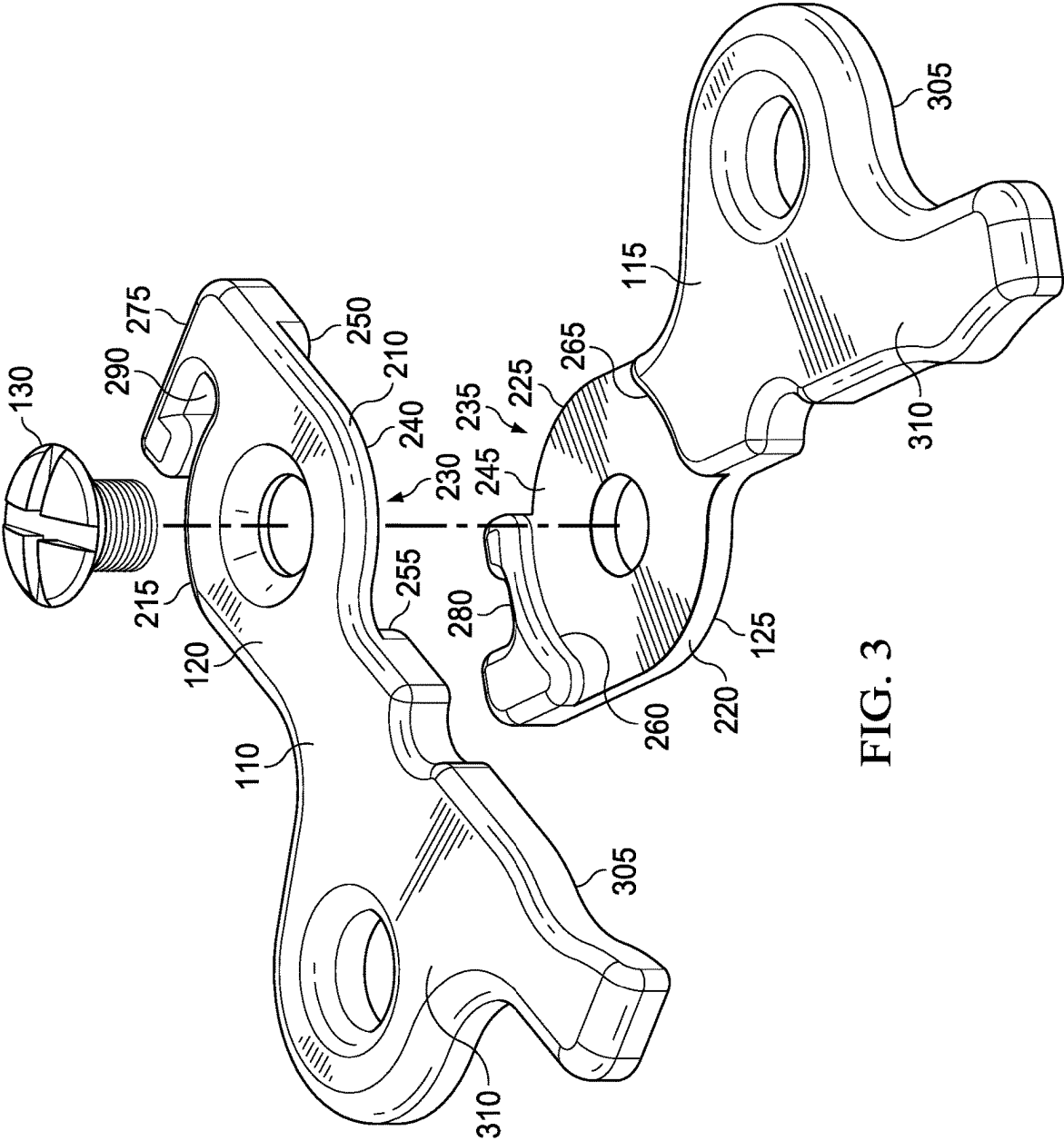


FIG. 3

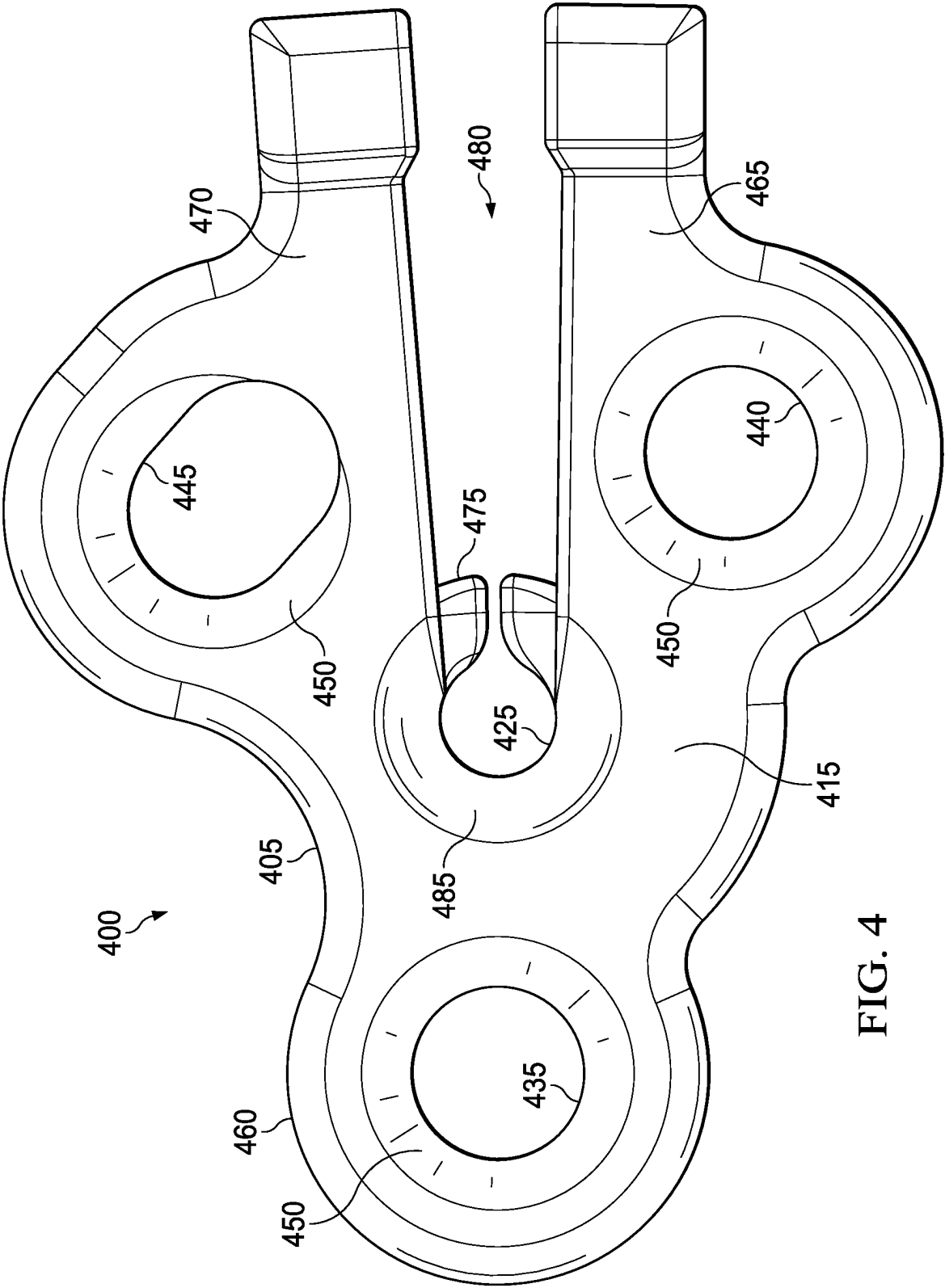


FIG. 4

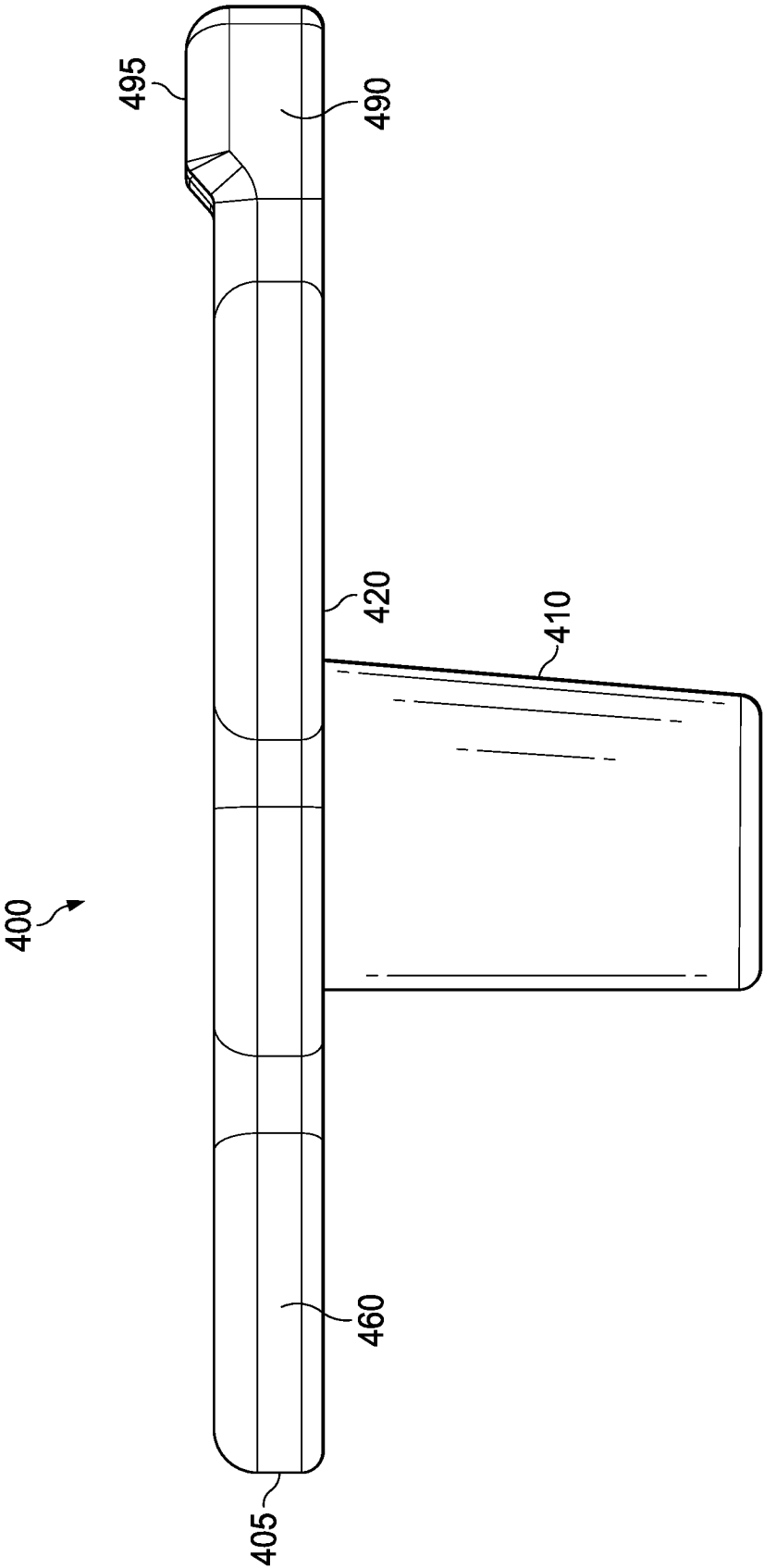


FIG. 5

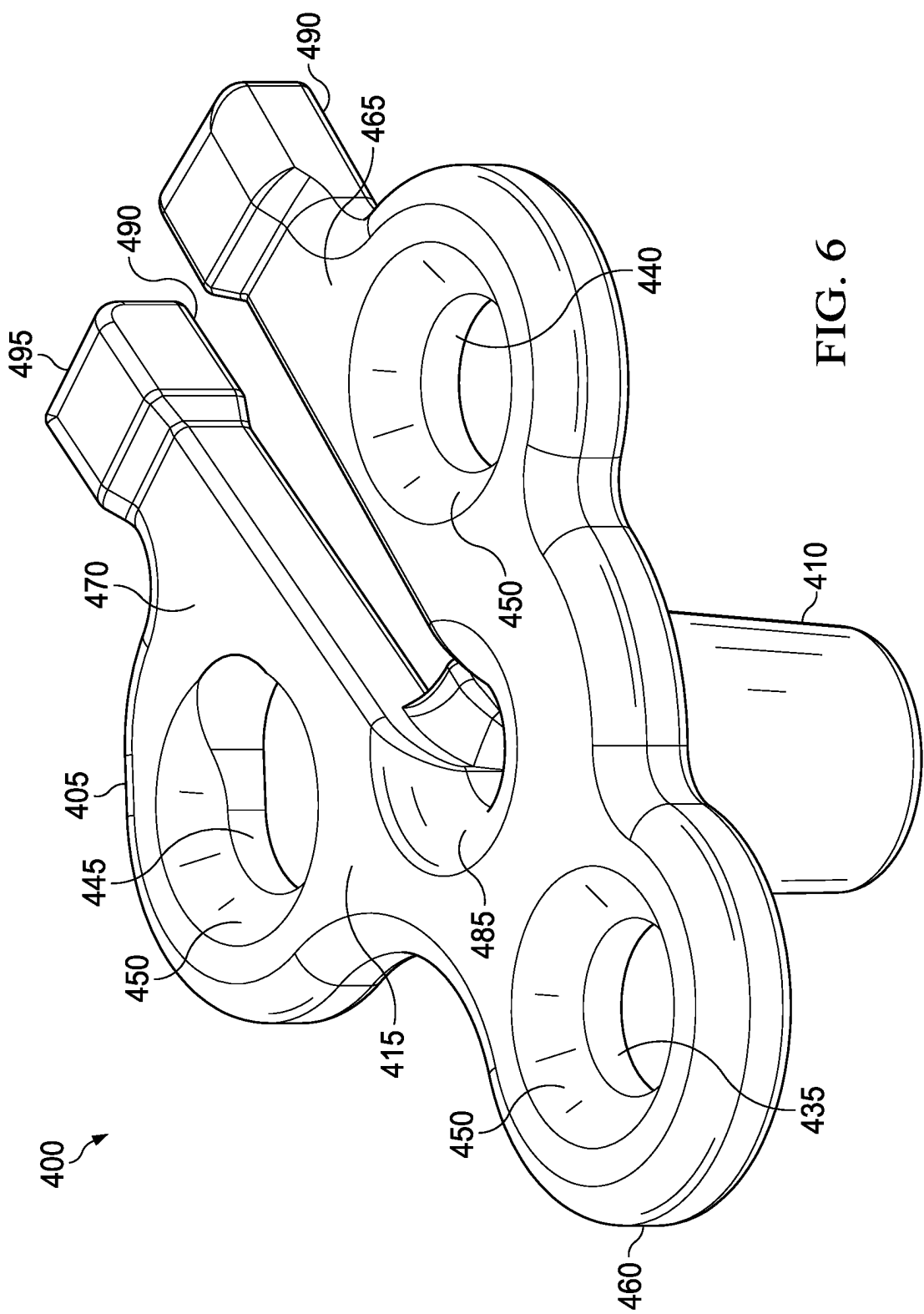
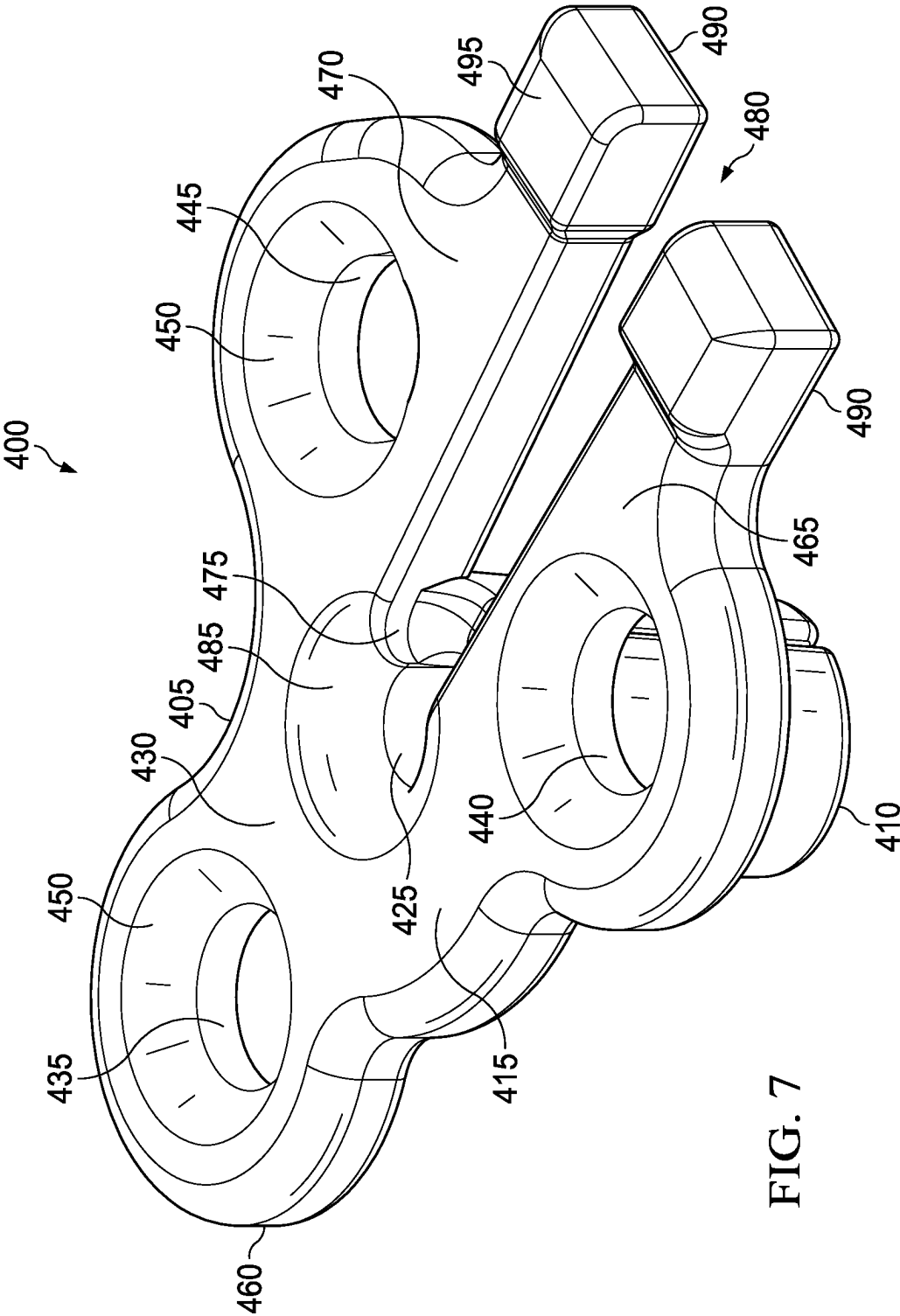


FIG. 6



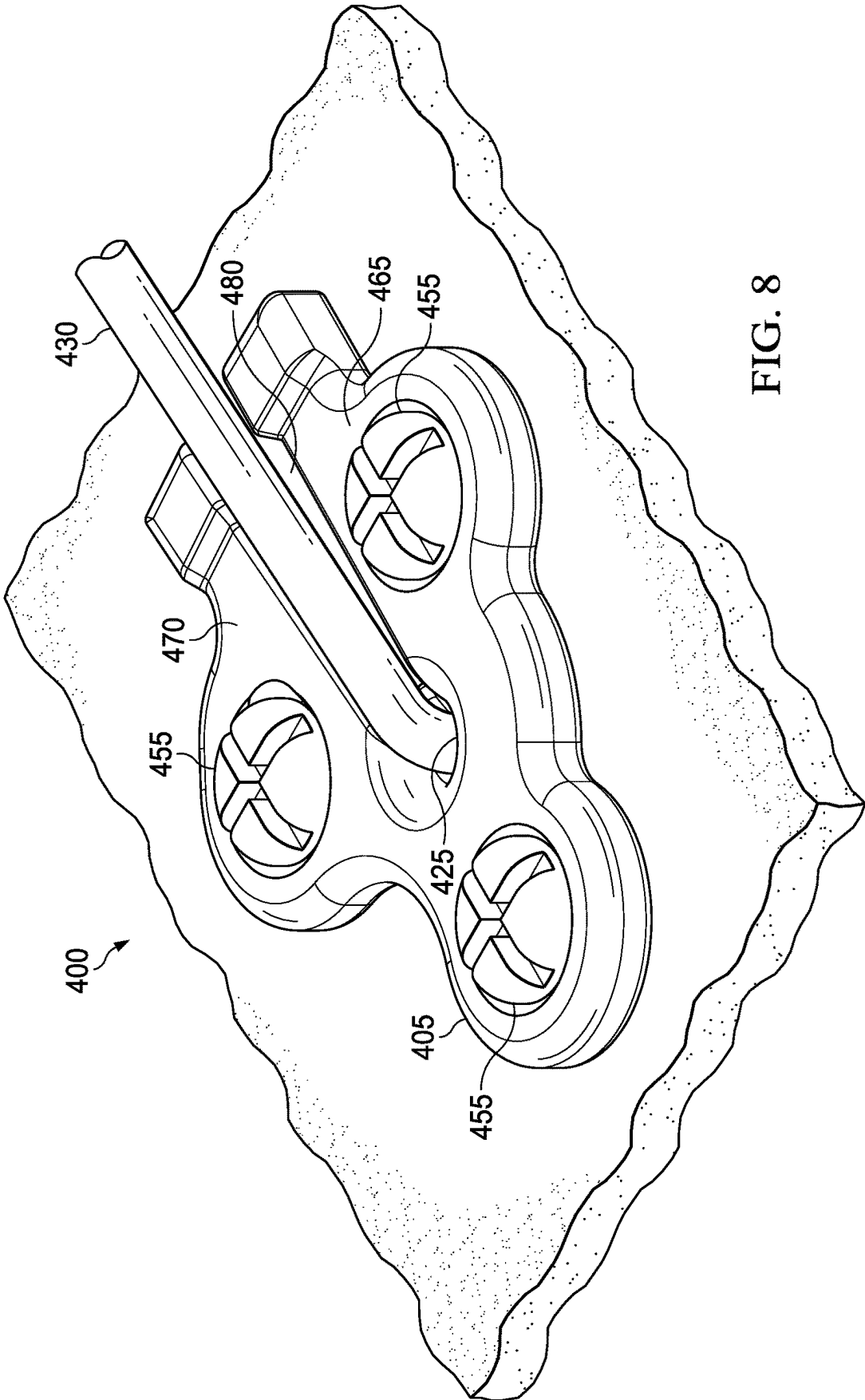
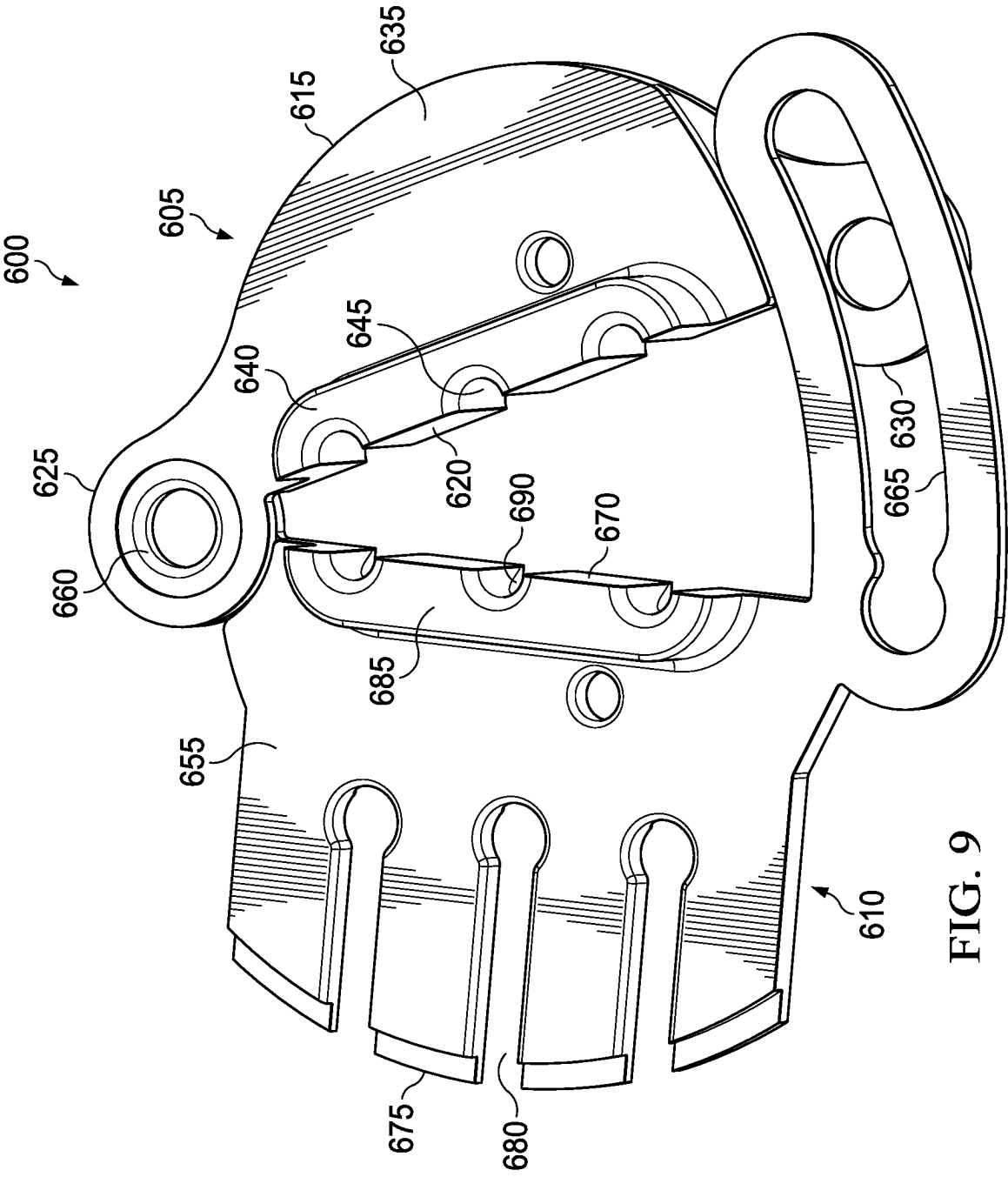
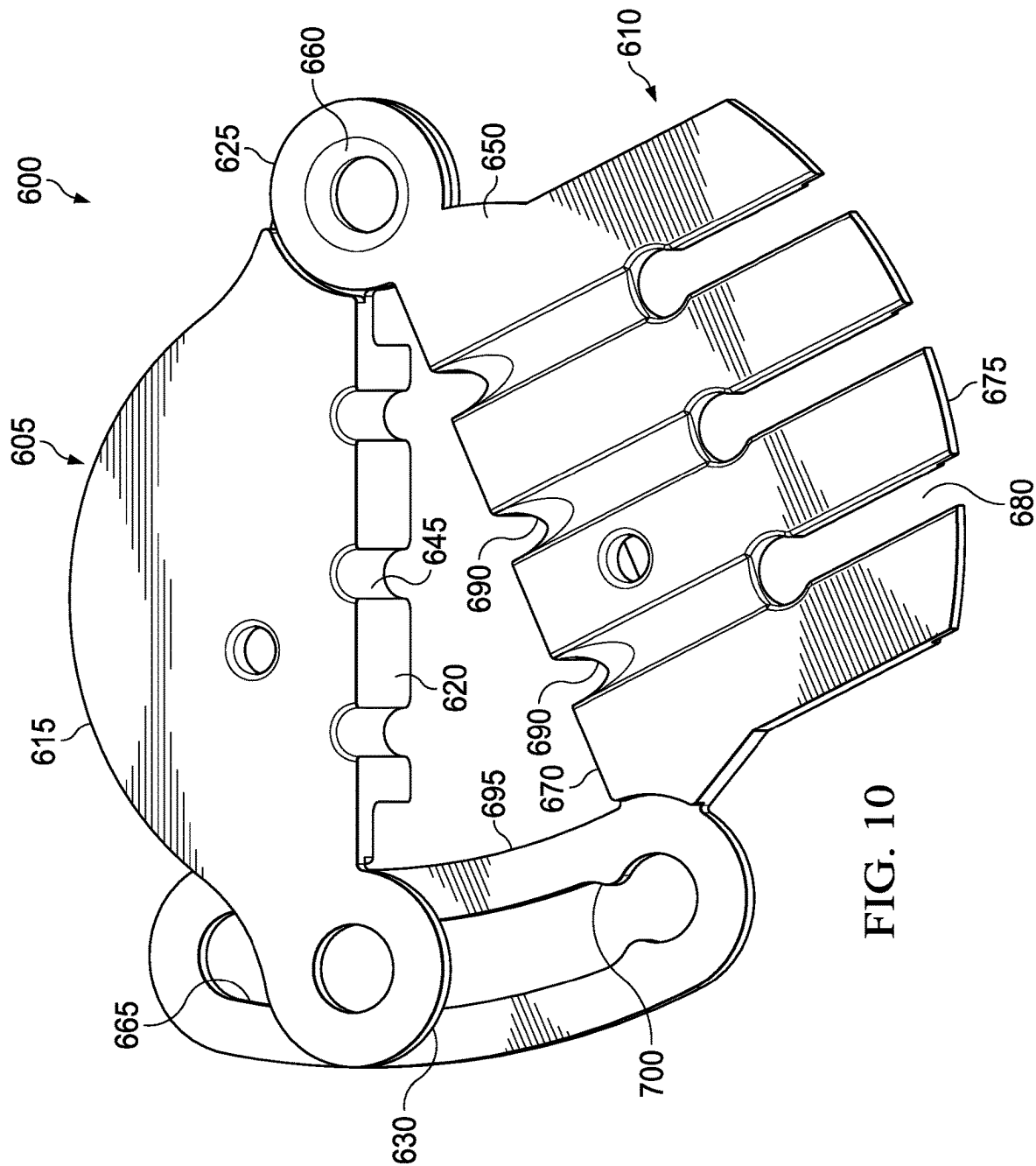


FIG. 8





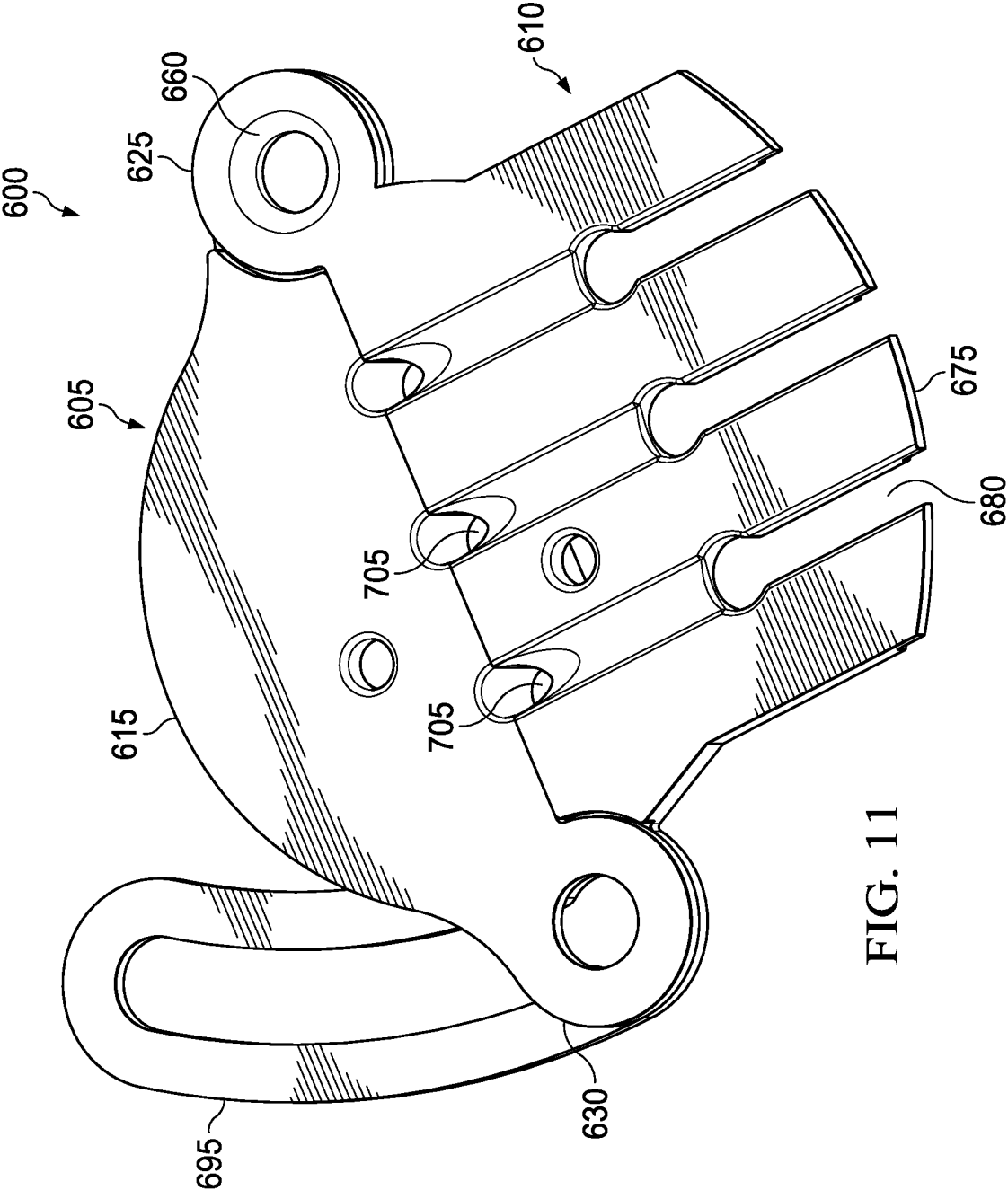


FIG. 11

METHODS AND APPARATUS FOR A LEAD FIXATION DEVICE

CROSS-REFERENCES TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 16/675,448, which claims the benefit of U.S. Provisional Patent Application No. 62/756,185, filed Nov. 6, 2018, and U.S. Provisional Patent Application No. 62/776,550, filed Dec. 7, 2018, and is a continuation-in-part of U.S. patent application Ser. No. 15/715,895 filed on Sep. 26, 2017, now patent Ser. No. 10,668,271 and incorporates the disclosure of each by reference. To the extent that the present disclosure conflicts with any referenced application, however, the present disclosure is to be given priority.

BACKGROUND OF THE TECHNOLOGY

Deep Brain Stimulation (DBS) is an established neuro-surgical treatment for multiple conditions including: Parkinson's disease, essential tremor, dystonia, and has been recently approved for medically refractory epilepsy. Responsive neural stimulation also uses implanted electrodes in the brain and is FDA approved. Additional probable indications likely to be approved in the future include: obsessive compulsive disorder, depression and even Alzheimer's disease. The number of permanently implanted brain electrodes continues to increase each year. Each brain electrode requires precise placement and permanent fixation, which allows for years of use. Generally the brain electrodes consist of DBS leads, which are implanted into the brain. The DBS leads are designed for durability and precise control of therapy.

DBS surgery typically involves placement of a single lead through a burr hole or a single twist drill opening, either unilaterally or bilaterally. A burr hole is a relatively small opening made in the bone of the skull to access the intracranial space (the space inside the skull). Burr holes are used to help relieve pressure on the brain when fluid, such as blood, builds up and starts to compress brain tissue. A burr hole is typically made by some form of spinning burr tool which can either be a manual, electric or pneumatic drill. The cutting bit is very much like a drill bit and allows the neurosurgeon to drill through the skull in a controlled fashion. The size of a typical burr hole is 14 mm. The size of a single twist drill opening may range from 2-5 mm in diameter for a single lead position. Other sizes may be contemplated based on the size of the lead fixation device.

In practice a lead is stereotactically implanted through a separate burr hole or single twist drill opening and then secured in place with a lead fixation device. A standard lead fixation device anchors the lead to the skull and covers the burr hole or single twist drill opening drilled through the skull. Standard lead fixation devices are designed for stable and secure fit against the skull and burr hole or single twist drill opening and to protect the lead/clip interface and create a smooth profile over the burr hole or single twist drill opening.

Unfortunately, all currently available lead fixation devices are somewhat cumbersome to use and result in a visible and palpable protrusion under the skin, which is usually bothersome to the patient. The visibility of the current lead fixation devices may also limit the use of DBS leads with bald patients, who may be concerned about the appearance

of "horns." One such example is the SureTek™ Burr Hole Cover Kit provided by Boston Scientific®.

SUMMARY OF THE TECHNOLOGY

A lead fixation device for attaching a deep brain stimulation lead within a burr hole in a human skull. The lead fixation device may comprise a mounting plate having first and second arms. The first arm may comprise a first flange, a first mounting hole, and a first inner wall. The second arm may comprise a second flange, a second mounting hole, and a second inner wall. The first and second arms may comprise first and second arcuate grooves located on the first and second inner walls. When the lead device in a closed position the first and second arcuate grooves form a guide channel and the first and second inner walls form an exit channel to receive the lead when the lead fixation device is in a closed position.

BRIEF DESCRIPTION OF THE DRAWINGS

A more complete understanding of the present technology may be derived by referring to the detailed description when considered in connection with the following illustrative figures. In the following figures, like reference numbers refer to similar elements and steps throughout the figures.

FIG. 1 representatively illustrates a top view of a lead fixation device in a closed position in accordance with various embodiments of the present technology;

FIG. 2 representatively illustrates a top view of the lead fixation device in an open position in accordance with various embodiments of the present technology;

FIG. 3 representatively illustrates an exploded, perspective view of the lead fixation device in an open position in accordance with various embodiments of the present technology;

FIG. 4 representatively illustrates a top view of an additional embodiment of a lead fixation device in an open position in accordance with various embodiments of the present technology;

FIG. 5 representatively illustrates a side view of the additional embodiment of a lead fixation device in an open position in accordance with various embodiments of the present technology;

FIG. 6 representatively illustrates a front side, perspective view of the additional embodiment of a lead fixation device in an open position in accordance with various embodiments of the present technology;

FIG. 7 representatively illustrates a rear side, perspective view of the additional embodiment the lead fixation device in an open position in accordance with various embodiments of the present technology;

FIG. 8 representatively illustrates a front side, perspective view the additional embodiment of the lead fixation device in an installed position in accordance with various embodiments of the present technology;

FIG. 9 representatively illustrates a bottom view of an additional embodiment of a lead fixation device in an open position in accordance with various embodiments of the present technology;

FIG. 10 representatively illustrates a top view of the additional embodiment of the lead fixation device in an open position in accordance with various embodiments of the present technology; and

FIG. 11 representatively illustrates a top view of the additional embodiment of the lead fixation device in a closed position in accordance with various embodiments of the present technology.

Elements and steps in the figures are illustrated for simplicity and clarity and have not necessarily been rendered according to any particular sequence. For example, steps that may be performed concurrently or in a different order are illustrated in the figures to help to improve understanding of embodiments of the present technology.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS

The present technology may be described in terms of functional block components and various processing steps. Such functional blocks may be realized by any number of components configured to perform the specified functions and achieve the various results. For example, the present technology may employ various types of materials, connectors, and the like for a lead fixation device, which may carry out a variety of operations suited to a specified application or environment. In addition, the present technology may be practiced in conjunction with any number of systems configured for operation with lead fixation devices, and the systems described are merely exemplary applications for the technology. Further, the present technology may employ any number of conventional techniques for utilizing and practicing lead fixation devices.

Methods and apparatus for providing a lead fixation device designed to minimize an invasive lead placement within the user and reduce visibility of the device to the public. Various representative implementations of the present technology may be applied to any system for minimizing invasive lead placement within the user and reduce visibility of the device to the public. The present technology may be utilized to minimize invasive lead placement within the user and reduce visibility of the device to the public.

The lead fixation devices described below may be comprised of any suitable materials. For example, the lead fixation device may comprise titanium, various plastic materials, or any other suitable medical grade material. In one embodiment, the lead fixation devices may be made of biocompatible Polyether ether ketone ("PEEK") material, which avoids wear and/or excessive compression of the lead. PEEK is a colorless organic thermoplastic polymer in the polyaryletherketone family, typically used in engineering applications. The PEEK material is constructed without sharp edges so that damage to the Deep Brain Stimulation ("DBS") lead does not occur.

Accordingly, a lead fixation device **100** is contemplated that is minimally invasive, smaller, and easier to install than any currently available system. The lead fixation device **100** allows the DBS lead to be placed through a small incision and bone opening than other current lead fixation devices. For example, the present lead fixation device **100** may be implanted in a bone opening of 3 mm. The smaller bone opening may be referred to as a twist drill hole or a biopsy drill hole. It should be understood that the lead fixation device **100** can be sized according to any suitable bone opening. In various embodiments, the lead fixation device **100** may be utilized with a single twist drill opening ranging from 2-5 mm in diameter. The lead fixation device **100** does not require any assembly or removal of the stereotactic at the time of lead fixation. In various embodiments, the height of the lead fixation device **100** may be range approximately 1 mm-3 mm. In one embodiment, the height of the lead fixation device **100** may be approximately 2 mm. In one embodiment, the height of the lead fixation device **100** may be approximately 1 mm. In one embodiment, the height of the lead fixation device **100** may be less than 1 mm.

In accordance with various embodiments, referring now to FIGS. 1-3, the lead fixation device **100** may comprise a mounting plate **105** comprising a first arm **110** and a second arm **115**. In one embodiment, the first and second arms **110**, **115** may comprise first and second flanges **120**, **125**. The first flange **120** of the first arm **110** may be coupled to the second flange **125** of the second arm **115**. In one embodiment, first and second flanges **120**, **125** of the first and second arms **110**, **115** are rotatable and coupled to each other. In one embodiment, first and second flanges **120**, **125** of the first and second arms **110**, **115** are press-fit together at the rotating hinge site with a circular flange collar. In other words one fitting inside in each arm resides within the other. In another embodiment, a coupling mechanism **130** is used to couple the first and second arms **110**, **115** to one another. The coupling mechanism **130** may be any screw, flange or coupling combination that couples the first and second arms **110**, **115** to one another. The second arm **115** is rotatable with regard to the first arm **110** such that when the two arms are pressed together, the lead fixation device **100** may secure the DBS lead to the patient/user.

In various embodiments, the first and second arms **110**, **115** of the mounting plate **105** may comprise a pair of mounting holes **135**, **140**, which extend therethrough. The first and second mounting holes **135**, **140** may be generally cylindrical. The first and second mounting holes **135**, **140** may be fixed once installed. Each of the first and second mounting holes **135**, **140** may contain an inwardly beveled sidewall **145** that receives a countersunk head of a mounting screw (not shown). Each of the first and second mounting holes **135**, **140** may be located at approximately the mid-point of the first and second arms **110**, **115**.

In one embodiment, the first and second arms **110**, **115** of the mounting plate **105** may each comprise an inner wall **150**, **155**. The inner walls **150**, **155** each may comprise a mating portion **160**, **165**, an arcuate groove **170**, **175**, and a longitudinal wall **180**, **185**. In one embodiment, the longitudinal walls **180**, **185**, may comprise a curved portion **190**, **195**.

FIG. 1 shows the lead fixation device **100** in closed position while FIGS. 2 and 3 show the lead fixation device **100** in an open position. In the closed position, mating portion **160**, **165** abut and the arcuate groove **170**, **175** form at least one guide channel **200** and the longitudinal walls **180**, **185** form an exit channel **205**. The guide channel **200** receives the DBS lead that exits the user's skull. In the closed position, the exit channel **205** formed within the space between the first and second arms **110**, **115** guides the DBS lead that exits the user's skull in a desired direction. In one embodiment, the first and second mounting holes **135**, **140** are located generally outwardly from the at least one guide channel **200**. The curved portions **190**, **195** and the longitudinal walls **180**, **185** when combined in the closed position form a "slalom slot" for press fitting the lead in its final position, after the lead stylet is removed, and to secure the lead against any delayed migration or external damage at the bending point where the lead exits the skull via the guide channel **200**.

Referring now to FIG. 3, the first and second flanges **120**, **125** will be discussed. In various embodiments, the first flange **120** may comprise first and second outer walls **210**, **215** and the second flange may comprise first and second outer walls **220**, **225**. The outer walls **210**, **215**, **220**, and **225** are generally curved and assist in the coupling and rotation of the first and second flanges **120**, **125**. In various embodiments, the first and second flanges **120**, **125** may comprise first and second mating portions **230**, **235**. In various

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embodiments, each of the first and second mating portions may comprise recessed areas **240, 245** where a portion of the first and second flanges **120, 125** has been removed. In various embodiments, the first flange **120** may comprise first and second collar walls **250, 255** and the second flange **125** may comprise first and second collar walls **260, 265**. The first and second collar walls **250, 255** of the first flange **120** are oriented opposite one another with the recessed area **240** locate there between. The first and second outer walls **210, 215** are located on the perimeter of the recessed area **240** between the first and second collar walls **250, 255**. The first and second collar walls **260, 265** of the second flange **125** are oriented opposite one another with the recessed area **245** locate there between. The first and second outer walls **220, 225** are located on the perimeter of the recessed area **245** between the first and second collar walls **260, 265**.

When assembled the recessed areas **240, 245** of the first and second mating portions **230, 235** contact one another; the outer walls **210, 215** of the first flange **120** are located adjacent the first and second collar walls **260, 265** of the second flange **125**; and the outer walls **220, 225** of the second flange **125** are located adjacent the first and second collar walls **250, 255** of the first flange **120**. This configuration allows the first arm **110** and second arm **115** to rotate with respect to each other and the height of the lead fixation device **100** to remain consistent.

In various embodiments, the lead fixation device **100** may comprise a locking mechanism **270**. The locking mechanism **270** may comprise first and second fingers **275, 280**. The first finger **275** depends from an upper surface of the first flange **120** of the first arm **110** and comprises a raised portion **285**. The first finger **275** is oriented such that a space **290** is formed between a lower surface **295** of the first finger **275** and the outer wall **215**. The second finger **280** depends from an upper surface of the second flange **125** of the second arm **115** and comprises a raised portion **300**. When the lead fixation device **100** is oriented in the closed position of FIG. **1**, the raised portion **300** is received within the space **290** and the raised portions **285, 300** lock the first and second arms **110, 115** of the lead fixation device **100**.

A lower surface **305** of the mounting plate **105** generally abuts the patient/user's skull while an upper surface **310** of the mounting plate **105** is generally flush with or sits on top of the patient/user's scalp. The lower surface **305** of the mounting plate **105** may be generally flat so that the lead fixation device is generally flush with the patient/user's scalp. This configuration provides a generally invisible or inconspicuous lead fixation device.

In operation, the DBS lead is inserted within a burr hole or a single twist drill opening in the user/patient's skull. While the lead fixation device **100** is oriented in the open position, shown in FIG. **2**, the DBS lead is placed within the guide channel **200**. The doctor may then press or pinch the arms **115, 120** toward one another and insert screws within the pair of mounting holes **135, 140** to fasten the lead fixation device **100** to the skull. The DBS lead is then placed within the exit channel **205** formed within the space between the first and second arms **110, 115**.

In accordance with various embodiments, referring now to FIGS. **4-8**, the lead fixation device **400** comprises a mounting plate **405** and an insertion collar **410**. The insertion collar **410** may be located at a central portion **415** of the mounting plate **405**. The insertion collar **410** depends downwardly from a lower surface **420** of the mounting plate **405**. The insertion collar **410** may be generally cylindrical and comprise an insertion hole **425** where a DBS lead **430** may be placed. In one embodiment, the insertion collar **410** may

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taper inwardly as it depends away from the lower surface **420**. The insertion hole **425** may extend though to an upper surface of the mounting plate **405**.

The mounting plate **405** may comprise a pair of mounting holes **435, 440** and a compression hole **445**, which extend therethrough. The first and second mounting holes **435, 440** may be generally cylindrical. The first and second mounting holes **435, 440** may be fixed once installed. Each of the first and second mounting holes **435, 440** contains an inwardly beveled sidewall **450** that receives a countersunk head of a mounting screw **455**. The compression hole **445** may comprise an elongate hole that allows for adjustment of the lead fixation device **400** during installation.

The first mounting hole **435** may be located on a flange **460** that extends from the central portion **415** of the mounting plate **405**. The second mounting hole **440** and the compression hole **445** reside on a pair of arms **465, 470** that extend from the central portion **415** opposite of the first mounting hole **435**. A space exists between the arms **465, 470** such that they can be pressed inwardly toward one another during installation. The arm **470** containing the compression hole **445** is rotatable with regard to the arm **465** containing the second mounting hole **440** such that when the two arms **465, 470** are pressed together, the space between the two arms **465, 470** creates a channel **480** to guide the DBS lead **430** exiting the patient/user.

The insertion hole **425** may comprise a cutout area **475** where a portion of a sidewall **485** of the insertion hole **425** is removed. The location of the cutout area **475** between the arms **465, 470** allows the arm **470** containing the compression hole **445** to move with regard to the arm **465** containing the second mounting hole **440**. The arms **465, 470** each contain a finger flange **490** with a raised portion **495**. The raised portions **495** allow the physician to press or pinch the arms **465, 470** toward one another during installation.

The lower surface **420** of the mounting plate **405** generally abuts the patient/user's skull and the insertion collar **410** may be inserted into the bone opening, burr hole or single twist drill opening of patient/user. The upper surface of the mounting plate **405** is generally flush with or sits on top of the patient/user's scalp. This configuration provides a generally invisible or inconspicuous lead fixation device.

FIGS. **4-7** show the lead fixation device **400** in an open position while FIG. **8** shows the lead fixation device **400** in an installed, closed position. In the closed position, the channel **480** formed within the space by the arms **465, 470** guides the DBS lead **430** that exits the user's skull in a desired direction.

In operation, the DBS lead **430** may be inserted within a burr hole or single twist drill opening in the user/patient's skull. While the lead fixation device **400** is oriented in the open position, shown in FIGS. **4-7**, the DBS lead **430** is placed within the insertion hole **425** located in the insertion collar **410**. The doctor may then fasten the lead fixation device **400** to the skull by inserting screws **455** within the first and second mounting holes **435, 440** and the compression hole **445**. The DBS lead may then be placed between the arms **465, 470**. The doctor may then press or pinch the raised portions **495** of the arms **465, 470** toward one another and tighten the screw **455** within the compression hole **445**. The DBS lead **430** will now reside within the channel **480** formed within the space between the arms **465, 470**.

Referring now to FIGS. **9-11**, in another embodiment, a low profile lead fixation device **600** is contemplated that is both faster and easier to use than any currently available system. The low profile lead fixation device **600** does not require any assembly or removal of the stereotactic at time

of lead fixation. The low profile lead fixation device **600** may be on the order of a millimeter in thickness, produces no visible bump, and avoids the need for countersinking the fixation device by excessive skull drilling at the time of lead placement. In various embodiments low profile lead fixation device **600** may range from approximately 2 mm to 0.5 mm in thickness. In one embodiment, low profile lead fixation device **600** may be 1 mm in thickness. In one embodiment, low profile lead fixation device **600** may be less than 1 mm in thickness.

In accordance with various embodiments, referring now to FIGS. 9-11, the low profile lead fixation device **600** will be discussed. The low profile lead fixation device **600** comprises a mounting plate having a first arm **605** and a second arm **610**. The first and second arms **605**, **610** are rotatable and coupled to each other. In one embodiment the first and second arms **605**, **610** are press-fit together at a rotating hinge site with a circular flange collar. In other words, one fitting inside in each arm resides within the other. The second arm **610** is rotatable with regard to the first arm **605** such that when the two arms are pressed together, the lead fixation device **600** may secure the DBS lead to the patient/user.

The first arm **605** may comprise a curved edge **615** and a substantially straight edge **620** located between a first flange **625** and a second flange **630**. A lower surface **635** of the first arm **605** may comprise a raised protrusion **640** residing along the substantially straight edge **620**. The raised protrusion **640** may comprise at least one arcuate groove **645**. In one embodiment the raised protrusion **640** may comprise 3 arcuate grooves **645**.

The second arm **610** may comprise an upper surface **650**, a lower surface **655**, a first flange **660**, and a ring slot **665**. The second arm **610** may comprise a substantially straight edge **670** and a curved edge **675** that reside between the first flange **660** and the ring slot **665**. The second arm **610** may comprise at least one exit channel **680**. In one embodiment, the second arm **610** may comprise 3 exit channels **680**. In one embodiment, each of the exit channels **680** may comprise a semi-circular hole. The semi-circular holes at each exit channel **680** are provided to allow easy insertion of the DBS lead wire for subsequent press fitting into the exit channel **680** for a secondary friction based fixation. The exit channels **680** are oriented generally perpendicular to the substantially straight edge **670** and extend from the straight edge **670** towards the curved edge **675**.

The lower surface **655** of the second arm **610** may comprise a raised protrusion **685** residing along the substantially straight edge **670**. The raised protrusion **685** may comprise at least one arcuate groove **690**. In one embodiment, the raised protrusion **685** may comprise 3 arcuate grooves **690**.

The lower surfaces **635**, **655** of the first and second arms **610**, **615** generally abut the patient/user's skull while the upper surface **650** is flush with patient/user's scalp. This configuration provides a generally invisible or inconspicuous lead fixation device.

The first flange **625** of the first arm **605** is coupled to the first flange **660** of the second arm **610**. The two arms **605**, **610** may be press fit together at the circular rotating hinge located at the first flanges **625**, **660**. The coupling of the first flanges **625**, **660** causes the second arm **610** to rotate with respect to the first arm **605**. The second flange **630** of the first arm **605** is aligned with the ring slot **665**. An inner wall **695** of the ring slot **665** abuts the curved edge **615** of the first arm **605** and slides along when the second arm **610** is rotated about the first arm **605**. The ring slot **665** may comprise a

pair of notches **700**, which are designed to engage the screw when the lead fixation device **600** is attached to the user.

FIGS. 9 and 10 show the lead fixation device **600** in an open position while FIG. 11 shows the lead fixation device **600** in a closed position. In the closed position, the raised protrusions **640**, **685** abut and the arcuate grooves **645**, **690** form at least one guide channel **705**. In one embodiment 3 guide channels **705** are formed. The guide channels **705** receive the DBS lead that exits the user's skull.

In operation, the DBS lead is inserted within a burr hole in the user/patient's skull. While the lead fixation device **600** is oriented in the open position, shown in FIGS. 9 and 10, the DBS lead is placed within the arcuate groove **645** located in the raised protrusion **640** of the first arm **605**. The second arm **610** is then rotated to the closed position, shown in FIG. 3, where the arcuate groove **690** located in the raised protrusion **685** of the second arm **610** forms the guide channel **705** and clamps the DBS lead to hold it in place. A stylet within the DBS lead may then be removed and the DBS lead may be pressed to fold over and fit within the exit channel **680** for a second point of fixation for added security. The doctor may then fasten the lead fixation device **600** to the skull by inserting screws within the first flanged connection **625**, **660** and the second flange **630** and ring slot **665** connection.

These and other embodiments for methods and apparatus for a lead fixation device may incorporate concepts, embodiments, and configurations as described above. The particular implementations shown and described are illustrative of the technology and its best mode and are not intended to otherwise limit the scope of the present technology in any way. Indeed, for the sake of brevity, conventional manufacturing, connection, preparation, and other functional aspects of the system may not be described in detail. Furthermore, the connecting lines shown in the various figures are intended to represent exemplary functional relationships and/or physical couplings between the various elements. Many alternative or additional functional relationships or physical connections may be present in a practical system.

The technology has been described with reference to specific exemplary embodiments. Various modifications and changes, however, may be made without departing from the scope of the present technology. The description and figures are to be regarded in an illustrative manner, rather than a restrictive one and all such modifications are intended to be included within the scope of the present technology. Accordingly, the scope of the technology should be determined by the generic embodiments described and their legal equivalents rather than by merely the specific examples described above. For example, the steps recited in any method or process embodiment may be executed in any order, unless otherwise expressly specified, and are not limited to the explicit order presented in the specific examples. Additionally, the components and/or elements recited in any apparatus embodiment may be assembled or otherwise operationally configured in a variety of permutations to produce substantially the same result as the present technology and are accordingly not limited to the specific configuration recited in the specific examples. Benefits, other advantages and solutions to problems have been described above with regard to particular embodiments; however, any benefit, advantage, solution to problems or any element that may cause any particular benefit, advantage or solution to occur or to become more pronounced are not to be construed as critical, required or essential features or components.

As used herein, the terms "comprises", "comprising", or any variation thereof, are intended to reference a non-

exclusive inclusion, such that a process, method, article, composition or apparatus that comprises a list of elements does not include only those elements recited, but may also include other elements not expressly listed or inherent to such process, method, article, composition or apparatus. Other combinations and/or modifications of the above-described structures, arrangements, applications, proportions, elements, materials or components used in the practice of the present technology, in addition to those not specifically recited, may be varied or otherwise particularly adapted to specific environments, manufacturing specifications, design parameters or other operating requirements without departing from the general principles of the same.

The present technology has been described above with reference to an exemplary embodiment. However, changes and modifications may be made to the exemplary embodiment without departing from the scope of the present technology. These and other changes or modifications are intended to be included within the scope of the present technology, as expressed in the following claims.

The invention claimed is:

1. A lead fixation device for attaching a deep brain stimulation ("DBS") lead within a burr hole in a human skull, the lead fixation device being movable from an open position to an operating position, comprising:

a mounting plate comprising:

- a mounting flange comprising a first static mounting hole configured to receive a mounting screw,
- a first arm comprising an inner wall and a second static mounting hole configured to receive a mounting screw, and
- a second arm comprising an inner wall and an adjustable mounting hole configured to receive a mounting screw, wherein the second arm is configured to rotate between the open position and the operating position when the mounting screw is received in the adjustable mounting hole,

wherein the second arm is movable to an operating position when the inner walls of the first and second arms are substantially parallel, and wherein the mounting screw for the adjustable mounting hole is tightened when the second arm is moved to the operating position.

2. The lead fixation device of claim 1, wherein the inner walls of the first and second arms form a channel to receive the DBS lead when the lead fixation device is in the operating position.

3. The lead fixation device of claim 2, wherein the second static mounting hole and the adjustable mounting hole are located outwardly of the channel.

4. The lead fixation device of claim 1, wherein the adjustable mounting hole comprises an elongated hole to provide for adjustment when the second arm is compressed toward the first arm when the lead fixation device is moved from the open position to the operating position.

5. The lead fixation device of claim 1, wherein:

- the second static mounting hole on the first arm is located at approximately a midpoint of the first arm, and
- the adjustable mounting hole is located at approximately a midpoint of the second arm.

6. The lead fixation device of claim 1, wherein the first and second arms each contain a raised flange at an end opposite the mounting flange, wherein the raised flange is configured to allow a user to move the second arm toward the first arm when the second arm is moved to the operating position.

7. The lead fixation device of claim 1, further comprising an insertion collar extending from a lower surface of the mounting plate and configured to be received within the burr hole in the human skull and comprising an insertion hole configured to receive the DBS lead.

8. The lead fixation device of claim 7, wherein the insertion collar and insertion hole are located inwardly of the first and second static mounting holes and the adjustable mounting hole.

9. The lead fixation device of claim 8, wherein an outer surface of the insertion collar tapers inwardly as it depends from the lower surface of the mounting plate.

10. The lead fixation device of claim 7, wherein the insertion hole comprises a cutout area where a portion of a sidewall of the insertion hole and insertion collar is removed.

11. A method of attaching the lead fixation device of claim 1 to a deep brain stimulation ("DBS") lead exiting a burr hole implanted in a human skull, comprising:

- placing the DBS lead within an insertion hole;
- inserting mounting screws within the first and second static mounting holes and the adjustable mounting hole to fasten the lead fixation device to the human skull;
- tightening the mounting screws within the first and second static mounting holes;
- pressing the second arm toward the first arm to form a channel while the mounting screw is inserted in the adjustable mounting hole;
- placing the DBS lead within the channel; and
- tightening the mounting screw within the adjustable mounting hole once the lead fixation device is in the operating position.

12. The method of attaching the lead fixation device of claim 11, wherein the inner walls of the first and second arms form the channel to receive the lead when the lead fixation device is in the operating position.

13. The method of attaching the lead fixation device of claim 11, wherein the adjustable mounting hole comprises an elongated hole to provide for adjustment when the second arm is compressed toward the first arm when the lead fixation device is moved from the open position to the operating position.

14. The method of attaching the lead fixation device of claim 11, wherein:

- the second static mounting hole on the first arm is located at approximately a midpoint of the first arm, and
- the adjustable mounting hole is located at approximately a midpoint of the second arm.

15. The method of attaching the lead fixation device of claim 14, wherein the second static mounting hole and the adjustable mounting hole are located outwardly of the channel.

16. The method of attaching the lead fixation device of claim 11, wherein the first and second arms each contain a raised flange at an end opposite the mounting flange, wherein the raised flange is configured to allow a user to move the second arm toward the first arm when the second arm is moved to the operating position.

17. The method of attaching the lead fixation device of claim 11, wherein the insertion hole is in an insertion collar.

18. The method of attaching the lead fixation device of claim 17, further comprising placing the insertion collar in the burr hole in the skull.

19. The method of attaching the lead fixation device of claim 17, wherein the insertion collar and insertion hole are located inwardly of the first and second static mounting holes and the adjustable mounting hole.

11**12**

20. The method of attaching the lead fixation device of claim **17**, wherein an outer surface of the insertion collar tapers inwardly as it depends from the lower surface of the mounting plate.

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